Spring boot virtual intership 6.0

Name : K Jithendar Reddy

Collage : KL University

Branch : CSE

Domin : Data Visualization

Project : Visualizing US Natural Disaster Declarations

Mentor : G.K.S Jyozteesh

Disaster Trends Analysis and Visualization

1. Project Description

The **Disaster Trends Analysis System** is a data analysis project that studies historical disaster declaration records to identify patterns and trends over time. The main goal of this project is to understand how disasters occur across different years and seasons using data visualization techniques. By analyzing the dataset, the project helps identify peak disaster periods and provides insights that can support disaster management and planning.

2. Dataset Used

The dataset used in this project contains records of disaster declarations including information such as disaster type, state, and declaration date. Each record represents a specific disaster event that occurred in a particular location. This dataset allows us to analyze disaster occurrences across different time periods and understand which disasters occur most frequently.

3. Environment Setup

The project was implemented using **Python** in a development environment such as **VS Code or Jupyter Notebook**. Several Python libraries were used for data processing and visualization, including **Pandas for data handling, NumPy for numerical operations, Matplotlib and Seaborn for creating graphs, and Plotly for interactive visualizations**. These tools help efficiently analyze large datasets and generate meaningful visual insights.

4. Data Exploration

The dataset was inspected using head(), info(), and describe() to understand structure and detect missing values.

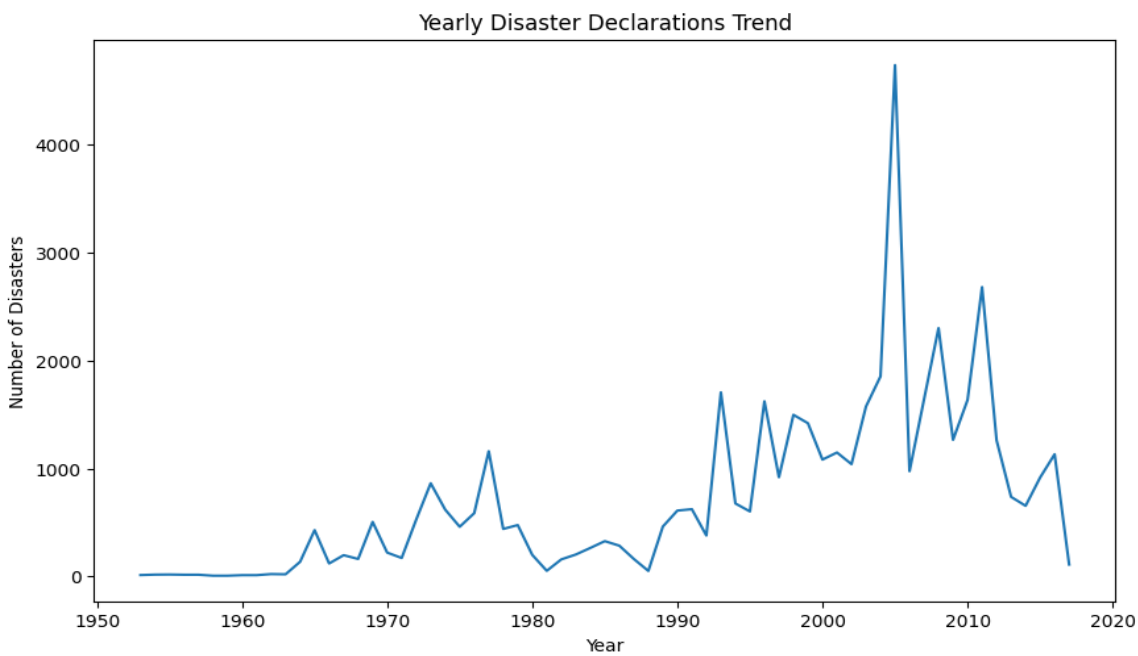
5. Data Preprocessing

n the preprocessing stage, the **Declaration Date** column was converted into datetime format to enable time-based analysis. From this date column, **year and month values were extracted** to study disaster trends across different years and seasonal patterns across months. This preprocessing step prepared the dataset for visualization and trend analysis.

6. Visualizations

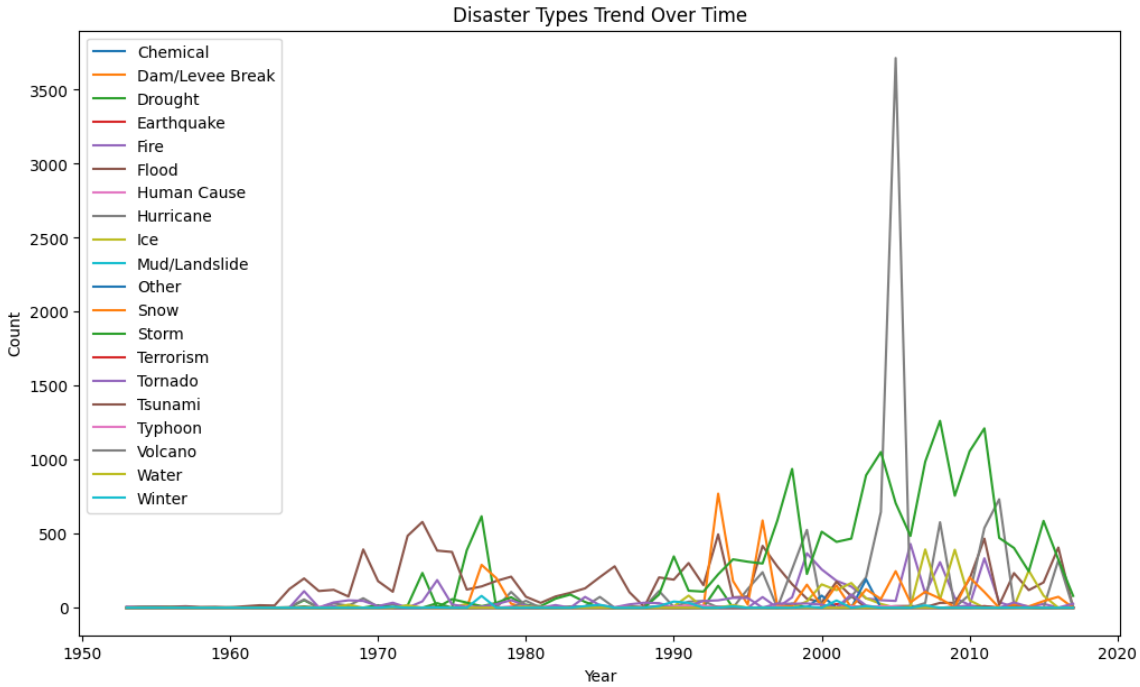
1 Yearly Disaster Trend

Figure 1 shows the number of disaster declarations each year.



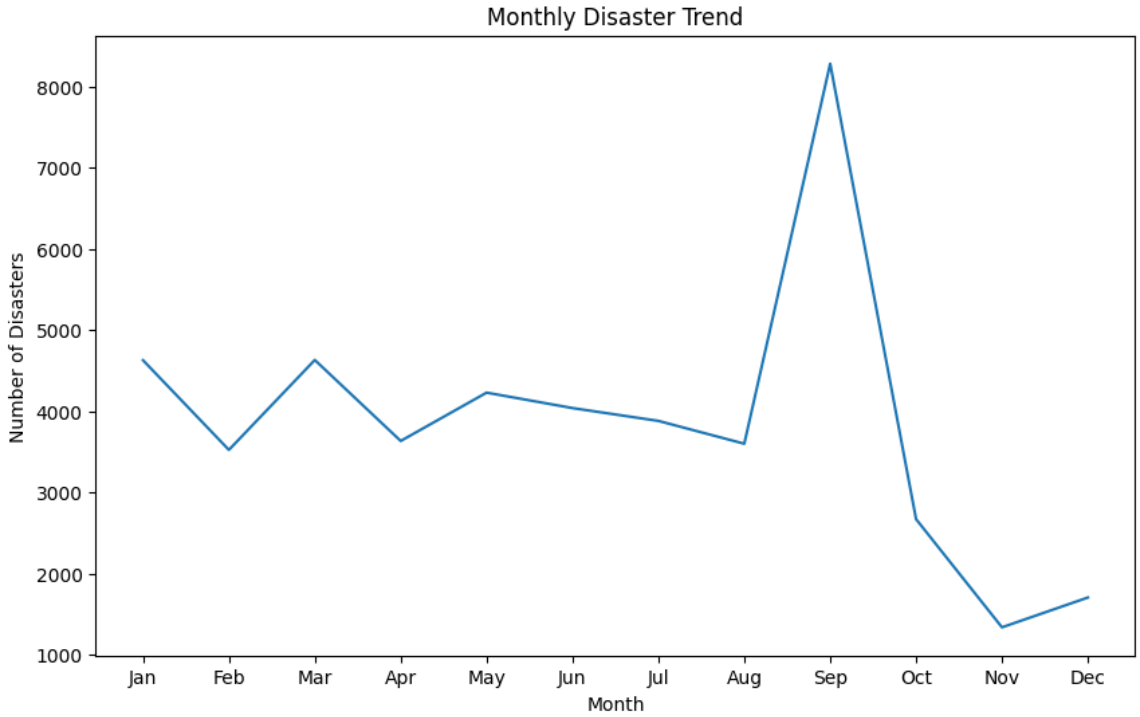
2 Disaster Type Trend

Figure 2 shows disasters categorized by incident type.



3 Monthly Disaster Trend

Figure 3 shows seasonal patterns of disasters by month.



7. Key Insights

Certain years experienced peak disasters. Seasonal patterns were visible in some months. Some disaster types occurred more frequently than others.

8. Conclusion

The project demonstrates how data visualization can reveal important trends in disaster occurrences and help improve disaster preparedness.